

Breaking the Speed Limit

Fast statistical models with Python 3.14, Numba, and JAX

Simulation gives us an answer key when real data has no ground truth.

Validate first. Then accelerate the bottleneck.

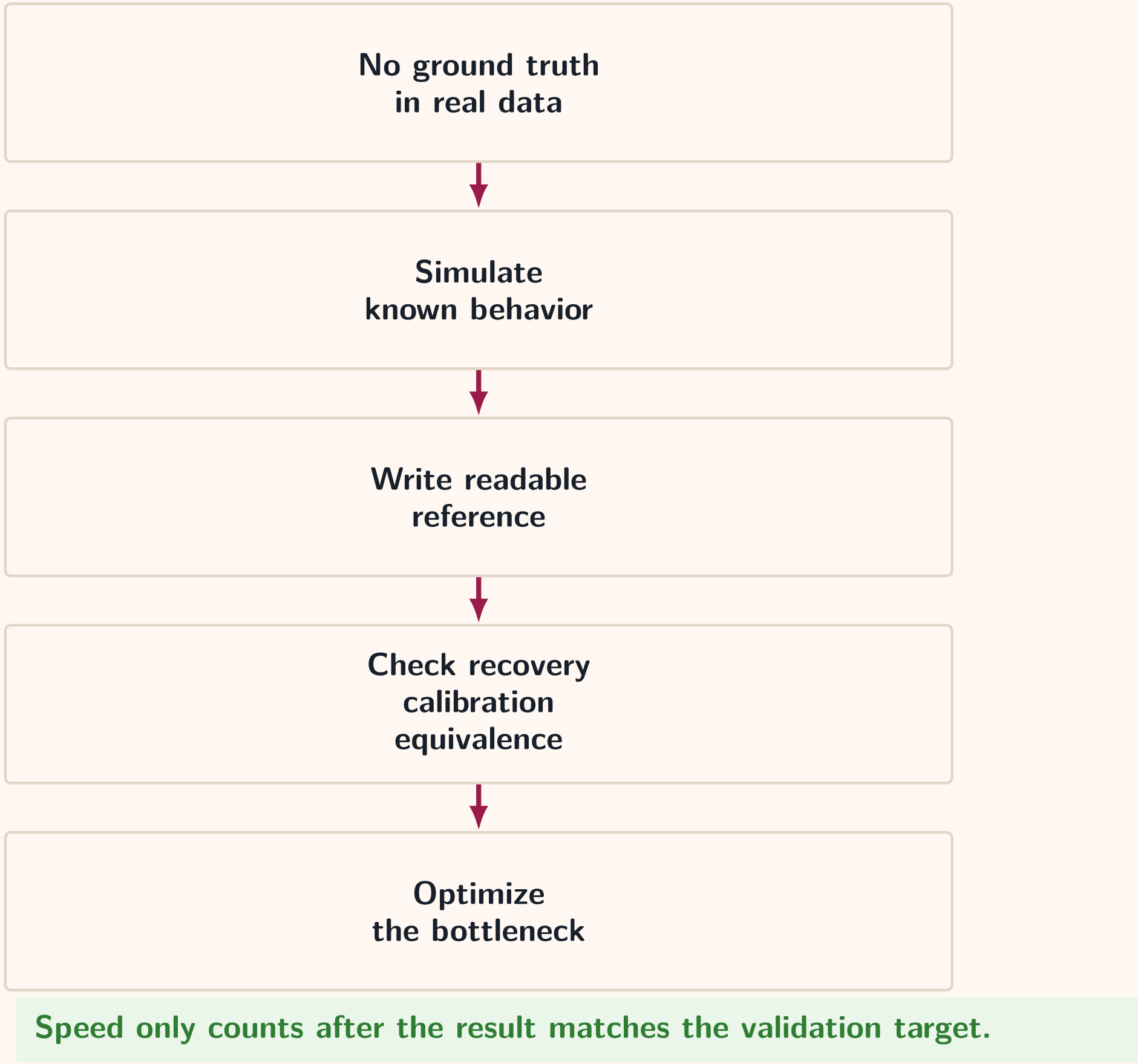
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github.com/LucaJiang/FastStatisticalModels4Python

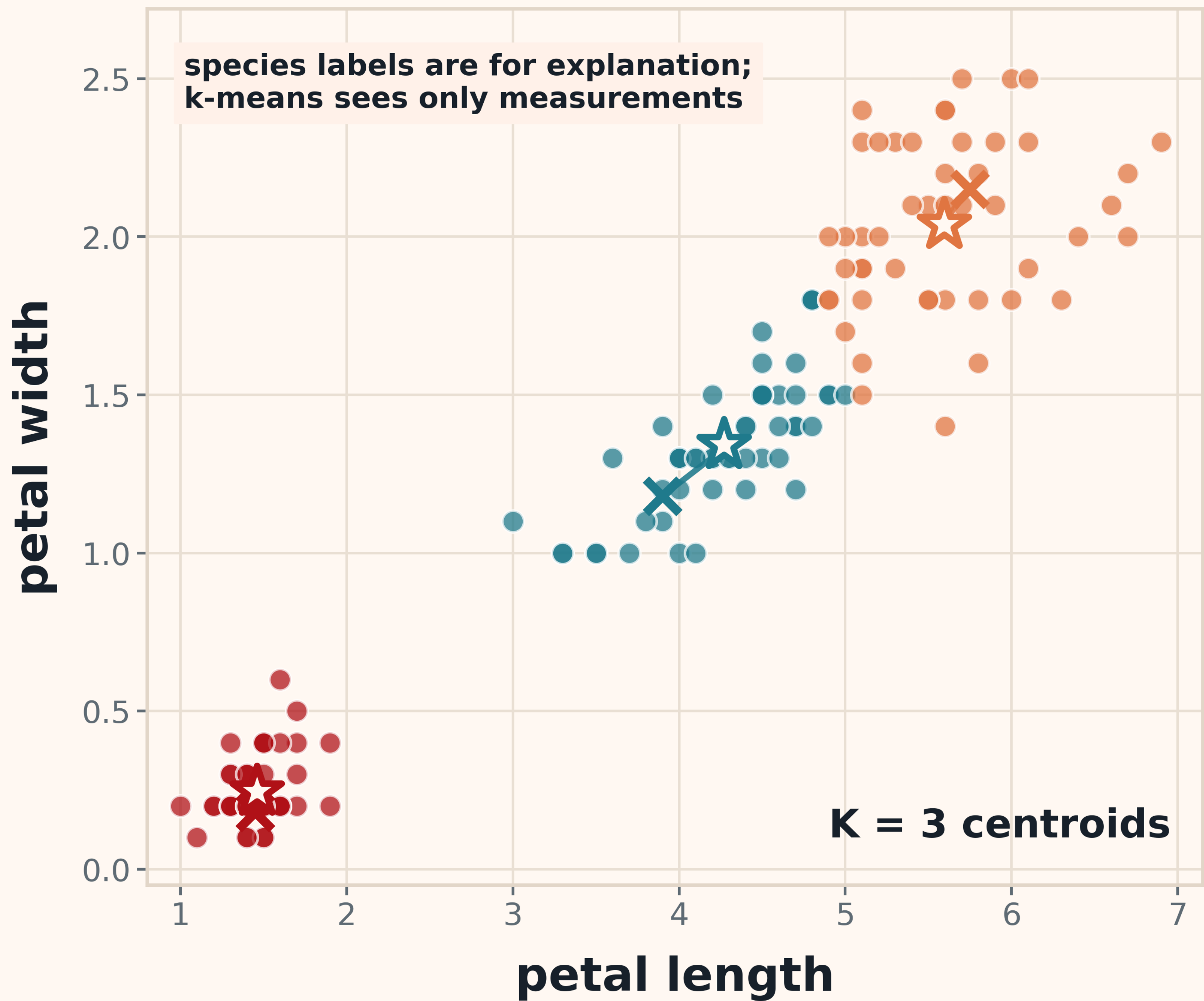
Why simulation?

An answer key for code



Workload 1: k-means

Iterative model fitting: assign → update → repeat

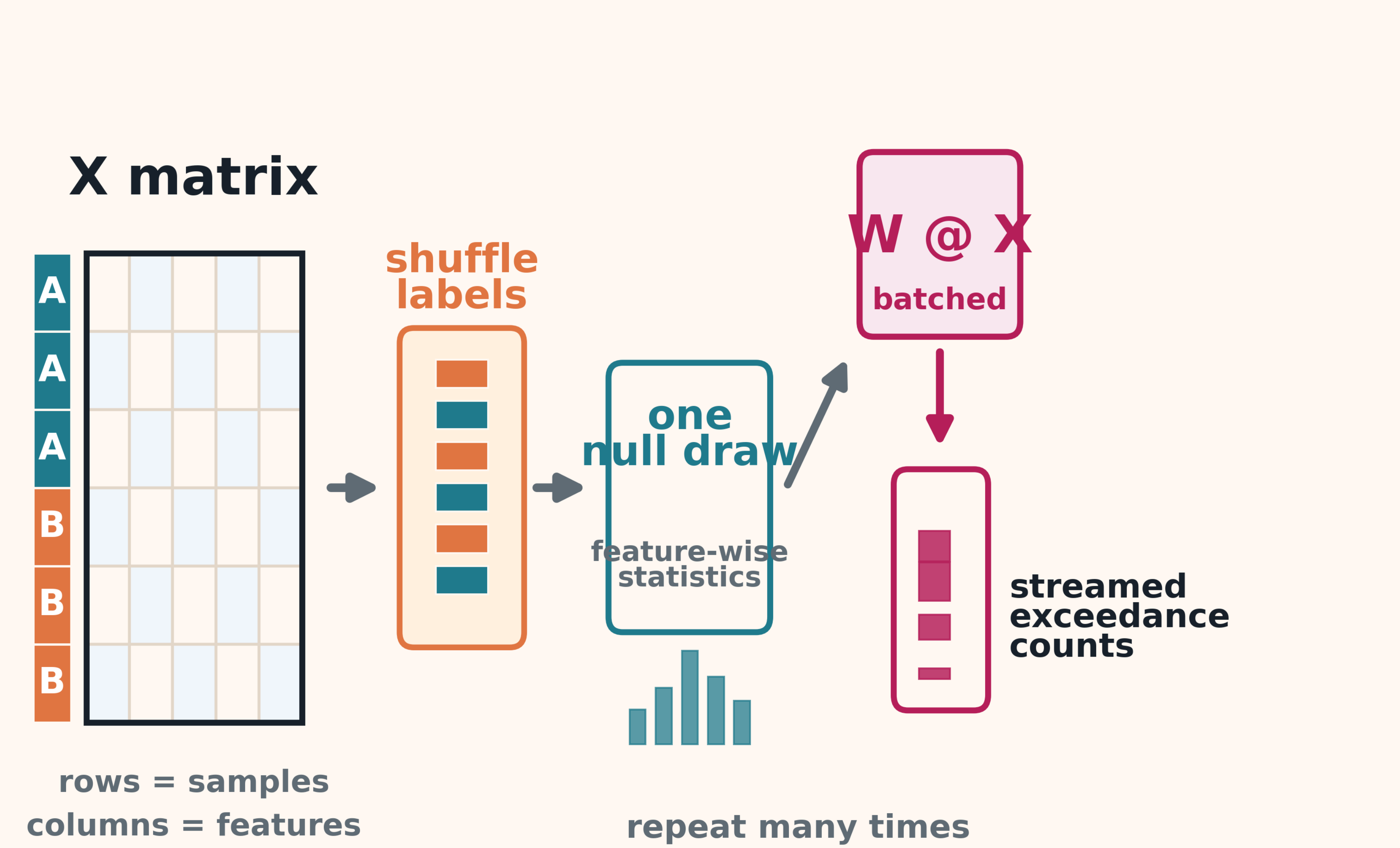


Same data and initialization.
Check recovery and final inertia.
Then compare implementations.

- Numba = loops
- NumPy/BLAS = distance algebra
- JAX/A100 = large regular arrays

Workload 2: permutation test

Resampling inference: shuffle → statistic → repeat



Same permutation stream.
Same p-value definition.
Check null calibration before timing.

- CPU wins when overhead dominates.
- A100 helps when $R \times p$ is large and the pipeline stays on device.

Simulate known behavior → Validate the result → Identify the bottleneck → Choose the smallest tool

- MacBook
validation and debugging
- Linux server CPU
parallelism and memory
- A100
batched device-resident workloads

Clear enough to trust. Fast enough to scale.